



Team title

Team Name: THE LEARNING LOOP

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 **COLLEGE NAME**

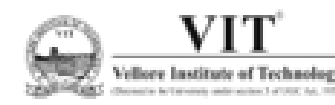
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Problem Statement Addressed

In SEM-based semiconductor inspection, a high-magnification reference image ($100\times$, $\sim 1\text{nm}/\text{pixel}$) needs to be located within a lower-magnification search image ($10\times$, $\sim 10\text{nm}/\text{pixel}$). Both images are 1000×1000 , and the goal is the match center in pixel coordinates. What makes this hard is the periodic nature of DRAM — the same cell pattern repeats every few pixels, so standard correlation-based matching frequently locks onto a visually identical but wrong location, causing errors of hundreds of pixels rather than a small offset.

- Repetitive DRAM structure causes false-positive matches on similar-looking cells
- Search image is noisier (dose variation, drift, charging effects)
- Scale isn't exactly $10\times$, and slight rotation (a few degrees) is common
- Manual alignment doesn't scale to production inspection volumes
- Need: one script, sub-pixel accurate (x, y) , works without retraining per image pair



Idea Description and Proposed Solution

KEY CONCEPT & APPROACH:

- Stage 1: correlation search across scales (9–11 $\times$) and rotations ($\pm 2^\circ$) to generate candidate matches, since exact scale and tilt between the two images are never precisely known Non-max suppression removes duplicate/overlapping peaks and narrows results down to top 20 candidates
- Stage 2: lightweight CNN ranks the 20 candidates, trained on real matcher errors (wrong DRAM cells) as hard negatives, so it learns exactly what fools correlation matching
- Stage 3: near-ties resolved by proximity to image center (our tie-break rule), then refined to sub-pixel accuracy via parabolic interpolation on the correlation peak

SOLUTION OVERVIEW:

- Single CLI tool — reference + search image $\rightarrow$ final (x, y) coordinates
- Physics-informed dataset generator creates realistic SEM training/test pairs
- Search images include realistic noise and acquisition variations
- No per-image retraining — frozen model at inference
- Pipeline is also adaptable to optical RGB inspection images



Innovation and Uniqueness

KEY INNOVATION

Periodic structures create ambiguity:

Standard correlation matching works well on unique features but fails on periodic structures because many locations score almost equally high.

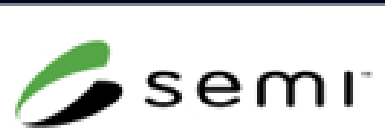
Our approach:

Instead of filtering these high-scoring candidates as noise, we treat them as useful information — they represent the ambiguity the matcher struggles with.

- Correlation generates the top candidate matches
- Correct match is present in ~90% of top-20 candidates
- CNN learns to resolve the ambiguity
- CNN chooses among candidates rather than detecting from scratch

COMPETITIVE ADVANTAGE

- 55–60% → improved matching compared with plain correlation within 5 px
- CNN is trained specifically on near-miss candidates
- <1 second per image pair on CPU
- Classical + CNN approach is efficient and interpretable
- Failures are understandable: wrong cell selected
- Training data follows real SEM imaging physics, not arbitrary synthetic shapes



Impact and Benefits

PRIMARY IMPACT

- Automates SEM alignment on highly repetitive DRAM patterns
- Handles 100×–10× repetitive structures without manual intervention
- No retraining required for new image pairs
- Runs entirely on CPU, with optional GPU acceleration
- Reduces inspection time, alignment errors, and manual rework

Quantifiable Outcomes

Split	Pairs	Within 5px	Within 2px	Within 1px	Median Error	Time
Eval	40	87.5%	85.0%	62.5%	0.66 px	~0.54 s
Test	250	85.6%	84.4%	68.4%	0.64 px	~0.54 s

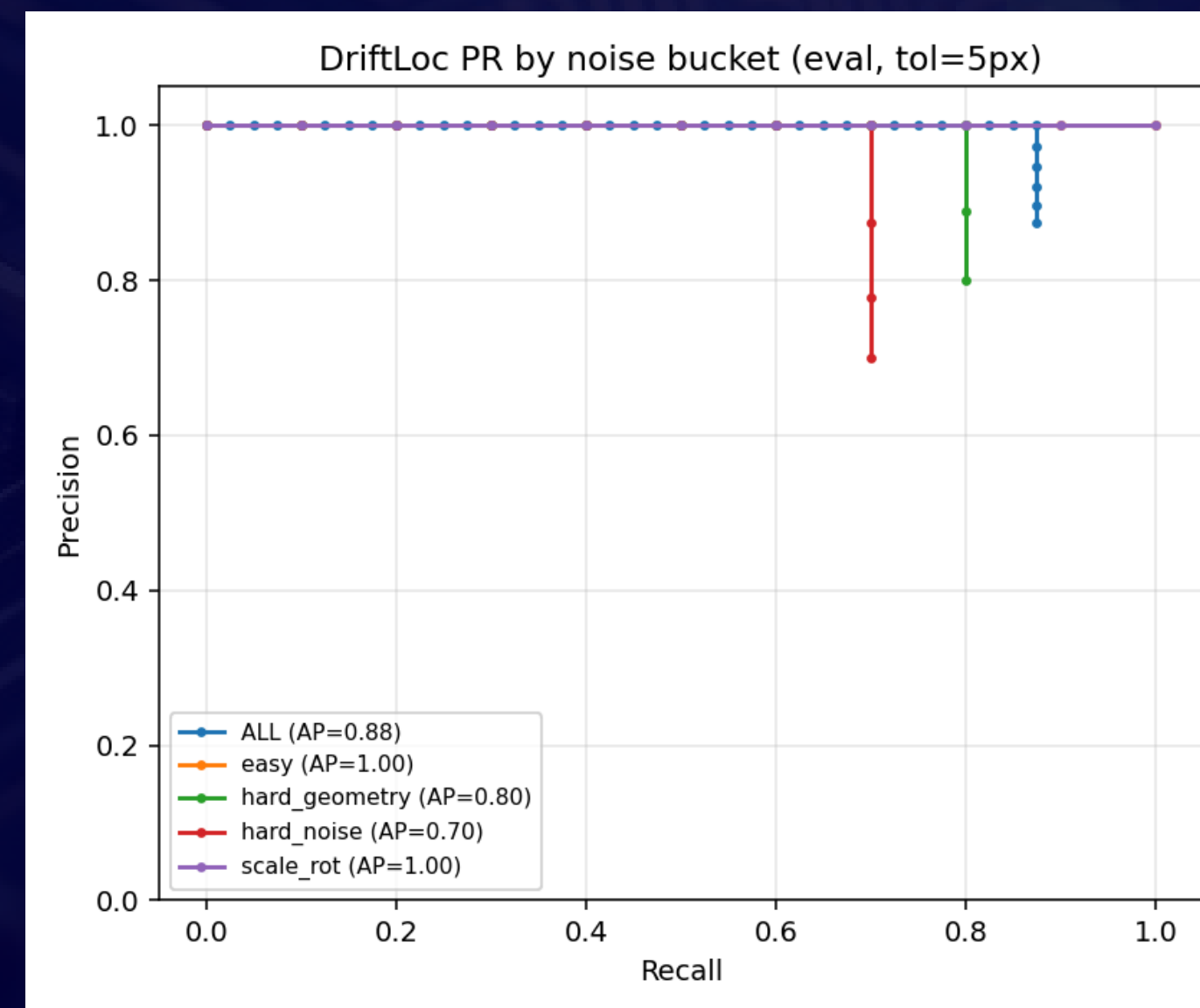
85.6% – Within 5 px
68.4% – Within 1 px
0.64 px – Median error



Technology & Feasibility/Methodology Used

The system is designed to be feasible without specialized SEM hardware for demonstration purposes, using a physics-informed simulation to approximate real imaging conditions closely enough for the model to generalize to actual inspection data.

- Pipeline: correlation search (25 scale × rotation combinations) → NMS top-20 candidates → CNN ranking → tie-break → sub-pixel refinement → final (x, y)
- Data generation: DRAM patterns built at 1nm resolution, cropped for reference, blurred/downsampled with realistic SEM noise (shot noise, detector noise, jitter, occasional charging) for search image
- 40-pair held-out evaluation set for testing
- Usage modes: single-pair CLI, batch mode via CSV, and a callable scoring function for integration
- Built on Python 3.10–3.12 with NumPy, OpenCV, and PyTorch
- Runs on a standard laptop CPU; GPU (CUDA/MPS) optional for speed





Research , Reference and Github,Video Link



GitHub Repository -- [Click Here](#)



Prototype / Simulation Video -- [Click here](#)



Research Background

SEM overlay: find a $100\times$ reference inside a $10\times$ search field. DRAM repeats, so correlation has many similar peaks and a global max often picks the wrong cell. We keep the top 20 ZNCC windows, rank them with a small CNN, apply the spec tie-break to $(500, 500)$, then sub-pixel refine. Generator uses published SEM artifacts (probe, shot noise, scan drift, charging), not fab layouts.



References & Citations

1. Applied Materials, Drift-Sense problem statement (SEMICON India 2026): $100\times$ vs $\sim 10\times$, 1000×1000 , center (x, y) , tie-break to $(500, 500)$.
2. Lewis, J. P. "Fast Normalized Cross-Correlation." Industrial Light & Magic, 1995. (ZNCC / stage-1)
3. Itoh, K. VLSI Memory Chip Design. Springer, 2001. ($6F^2$ DRAM pitch ratios)
4. Reimer, L. Scanning Electron Microscopy. Springer. (probe / astigmatism)
5. Jones, L. & Nellist, P. D. Microscopy and Microanalysis, 19(4), 2013. (scan drift / jitter)